

A counterexample to the fivefold sumset inequality

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Abstract

We give finite sets of integers A, B satisfying $|A - B|^5 > |A + B|^5 |5B|$, where $5B$ is the fivefold sumset of B . The sets are base-193 encodings of nonnegative integer vectors in dimension 128, with coordinate sums bounded by 160 and 32, respectively. Stars and bars counts the sumsets, and a sum of 33 binomial terms counts the difference set. The strict comparison uses integer arithmetic and is independently reproduced by counting both positive and negative supports. A second construction, using a fixed-weight binary digit set, gives a finite proof by type counting and positive generating polynomials; its strict comparison follows from small rational inequalities and the binomial theorem.

Keywords. sumset; difference set; integer simplex; type counting; exact certificate. **MSC 2020.** 11B75; 05A16.

1 The inequality and the construction

For finite nonempty subsets A, B of an abelian group, write $A \pm B = \{a \pm b : a \in A, b \in B\}$ and $hB = \{b_1 + \dots + b_h : b_1, \dots, b_h \in B\}$. Gyarmati, Hennecart and Ruzsa [1, equation (9), p. 177] asked for which integers h the inequality

$$|A - B| \leq |A + B| |hB|^{1/h} \tag{1}$$

holds for all such sets. Their discussion on p. 182 gives a counterexample for $h = 6$ and leaves $h = 4, 5$ unresolved. The result below gives an explicit counterexample to the $h = 5$ statement in that text; no priority or current-status claim is made.

For integers $m \geq 1$ and $r \geq 0$, define the integer simplex

$$V(m, r) = \left\{ x \in \mathbb{Z}_{\geq 0}^m : \sum_{i=0}^{m-1} x_i \leq r \right\}.$$

Its size is $\binom{m+r}{m}$. The lattice-simplex construction and its equal-radius difference count appear in Hennecart, Robert and Yudin [3, Section 3]; Gyarmati, Hennecart and Ruzsa use this family for their $h = 6$ example [1, p. 182]. Glasscock [2, Lemma B.1, pp. 37–38] gives the exact count for unequal radii, including the negative-support proof reproduced below. His Question 8 on p. 28 also leaves $h = 4, 5$ unresolved. Our example specializes that known formula to $(m, a, b) = (128, 160, 32)$ and verifies the strict fivefold comparison.

*Developed with AI research agents. The simplex construction is accompanied by two exact integer count formulas and a separate verifier; the second construction has a self-contained finite proof.

Theorem 1. Let $\phi : \mathbb{Z}^{128} \rightarrow \mathbb{Z}$ be the additive map

$$\phi(x) = \sum_{i=0}^{127} 193^i x_i.$$

The finite integer sets

$$A = \phi(V(128, 160)), \quad B = \phi(V(128, 32))$$

satisfy

$$|A - B|^5 > |A + B|^5 |5B|. \quad (2)$$

2 Counting the vector sets

We recall the unequal-radius count from [2, Lemma B.1] and give a second expression for its arithmetic verification.

Lemma 2. For integers $m \geq 1$ and $a, b \geq 0$, let $X = V(m, a)$ and $Y = V(m, b)$. Then

$$|X + Y| = \binom{m+a+b}{m}, \quad |5Y| = \binom{m+5b}{m}, \quad (3)$$

$$|X - Y| = \sum_{j=0}^{\min(m,b)} \binom{m}{j} \binom{b}{j} \binom{m-j+a}{m-j}. \quad (4)$$

The difference count also equals

$$\sum_{j=0}^{\min(m,b)} \sum_{k=0}^{\min(m-j,a)} \binom{m}{j} \binom{m-j}{k} \binom{b}{j} \binom{a}{k}. \quad (5)$$

Proof. One has $V(m, a) + V(m, b) = V(m, a+b)$. To obtain the reverse inclusion, allocate up to a units of any vector's coordinate mass to the first summand; the remaining mass is at most b . Iterating this identity gives $5Y = V(m, 5b)$. Stars and bars proves (3).

For $z \in \mathbb{Z}^m$, put $(z_+)_i = \max(z_i, 0)$ and $(z_-)_i = \max(-z_i, 0)$. Then

$$z \in X - Y \iff \sum_i (z_+)_i \leq a \text{ and } \sum_i (z_-)_i \leq b.$$

Necessity follows from any representation $z = x - y$; sufficiency follows by taking $x = z_+$ and $y = z_-$. If z has exactly j negative coordinates, their positions can be chosen in $\binom{m}{j}$ ways. Their positive magnitudes have total at most b , giving $\binom{b}{j}$ choices. The other $m-j$ coordinates are nonnegative with total at most a , giving $\binom{m-j+a}{m-j}$ choices. This proves (4), including the case $j = 0$.

Alternatively, choose the j negative positions and then the k positive positions. Their nonzero magnitudes can be filled in $\binom{b}{j}$ and $\binom{a}{k}$ ways. Summing over the disjoint supports proves (5). $\square$

For the parameters of Theorem 1, set

$$D = \sum_{j=0}^{32} \binom{128}{j} \binom{32}{j} \binom{288-j}{128-j}, \quad (6)$$

$$S = \binom{320}{128}, \quad F = \binom{288}{128}. \quad (7)$$

These integers satisfy $D^5 > S^5 F$. The arithmetic can be checked with the following standalone Python code, using integer operations only:

```
from math import comb
D = sum(comb(128,j)*comb(32,j)*comb(288-j,128-j)
        for j in range(33))
S, F = comb(320,128), comb(288,128)
if D**5 <= S**5*F:
    raise ArithmeticError("fivefold comparison failed")
print("D^5 > S^5 F")
```

The accompanying verifier also evaluates (5) with factorials instead of the binomial-coefficient routine and checks the resulting integers against the stored values. Its small-set controls enumerate the sets themselves.

3 Encoding as integer sets

Proof of Theorem 1. Write $X = V(128, 160)$ and $Y = V(128, 32)$. Every coordinate of $X + Y$ lies in $[0, 192]$, and every coordinate of $5Y$ lies in $[0, 160]$. Base-193 uniqueness therefore makes ϕ injective on both sets.

Every coordinate of $X - Y$ lies in $[-32, 160]$. Adding 32 to each coordinate gives digits in $[0, 192]$ and adds the same integer $32 \sum_{i=0}^{127} 193^i$ to every image. Thus ϕ is injective on $X - Y$ as well. Additivity now gives

$$|A - B| = D, \quad |A + B| = S, \quad |5B| = F.$$

The integer comparison following (7) proves (2). □

4 A second construction by finite type counting

The same fivefold reversal also has a construction whose strict comparison uses only small rational inequalities and the binomial theorem. Its dimension is larger than that of Theorem 1. An asymptotic version follows from Lau and Nair's typical-set theorem [5, Theorem 10, Appendix C], applied with probability laws

$$p = \frac{(144, 64, 32, 16, 8, 4, 2, 1)}{271}, \quad q = \frac{(255, 16)}{271}$$

on $\{0, \dots, 7\}$ and $\{0, 1\}$, respectively, together with elementary generating-polynomial bounds. One binary typical set supplies all five summands. The finite type-counting proof below gives the stated dimension directly. Matolcsi, Ruzsa, Shakan and Zhelezov [4, Section 8] develop the related weighted max-convolution framework.

Theorem 3. *Set $k = 2048$, $n = 271k = 555008$, and define*

$$X = \left\{ x \in \{0, \dots, 7\}^n : \sum_{i=0}^{n-1} x_i = 247k \right\}, \quad Y = \left\{ y \in \{0, 1\}^n : \sum_{i=0}^{n-1} y_i = 16k \right\}.$$

For $\psi(v) = \sum_{i=0}^{n-1} 9^i v_i$, the finite nonempty integer sets $A = \psi(X)$ and $B = \psi(Y)$ satisfy $|A - B|^5 > |A + B|^5 |5B|$.

Proof. Consider all words with digit multiplicities

$$(m_{-1}, m_0, m_1, \dots, m_7) = k(16, 128, 64, 32, 16, 8, 4, 2, 1).$$

Each belongs to $X - Y$: at a digit -1 take $(x_i, y_i) = (0, 1)$; at every other digit d take $(x_i, y_i) = (d, 0)$. The coordinate sums are $16k$ for y and $(64 + 2 \cdot 32 + 3 \cdot 16 + 4 \cdot 8 + 5 \cdot 4 + 6 \cdot 2 + 7)k = 247k$ for x . These words are distinct difference vectors, so

$$|X - Y| \geq \frac{n!}{\prod_{d=-1}^7 m_d!} \geq \frac{Q^n 2^{295k}}{(n+1)^9}, \quad Q = \frac{271}{128}. \quad (8)$$

To see the second inequality directly, put $p_d = m_d/n$. In the multinomial distribution with probabilities p_d , the displayed type is modal: for each positive integer m , the sequence $m^j/j!$ is maximal at $j = m$, and a type probability is a common factor times $\prod_d m_d^{j_d}/j_d!$. There are at most $(n+1)^9$ types. Thus the displayed type has probability at least $(n+1)^{-9}$. Substituting $\prod_d p_d^{m_d} = Q^{-n} 2^{-295k}$ proves the bound.

Every $X + Y$ output has digits in $\{0, \dots, 8\}$ and total $263k$. Every $5Y$ output has digits in $\{0, \dots, 5\}$ and total $80k$. For a polynomial with nonnegative coefficients, its coefficient of z^r is at most its value at any $z > 0$, divided by z^r . Apply this to $(1 + z + \dots + z^8)^n$ at $z = 1/2$ and to $(1 + z + \dots + z^5)^n$ at $z = 1/4$. This gives

$$|X + Y| \leq P^n 2^{263k}, \quad |5Y| \leq T^n 2^{160k}, \quad P = \frac{511}{256}, \quad T = \frac{1365}{1024}. \quad (9)$$

The coefficient bounds count all words with the prescribed digit totals; they require no assertion that every such word is attainable. Since $5 \cdot 295 = 5 \cdot 263 + 160$, the bounds imply

$$\frac{|X - Y|^5}{|X + Y|^5 |5Y|} \geq \frac{R^n}{(n+1)^{45}}, \quad R = \frac{(542/511)^5}{1365/1024}.$$

Small rational comparisons suffice:

$$\frac{542}{511} > \frac{53}{50}, \quad \left(\frac{53}{50}\right)^5 > 1 + 5\frac{3}{50} + 10\left(\frac{3}{50}\right)^2 = \frac{167}{125}, \quad \frac{171008}{170625} > \frac{513}{512}.$$

The last fraction before $513/512$ is $(167/125)/(1365/1024)$; its cross-product difference is 25471. Hence $R > 1 + 1/512$. As $n = 512 \cdot 1084$ and $n + 1 < 2^{20}$, the binomial theorem yields

$$R^n > ((1 + 1/512)^{512})^{1084} > 2^{1084} > 2^{900} > (n+1)^{45}.$$

This proves the strict vector-set inequality.

Finally, the digits of $X - Y$, $X + Y$ and $5Y$ lie in $[-1, 7]$, $[0, 8]$ and $[0, 5]$, respectively. Base 9 is injective on each of these vector sets: at a largest differing position j , the leading contribution has magnitude at least 9^j , while all lower contributions have total magnitude at most $8 \sum_{i < j} 9^i = 9^j - 1$. Additivity of ψ therefore preserves all three cardinalities and proves the integer-set assertion. In particular, B consists of the $\binom{n}{16k}$ base-9 words with exactly $16k$ digits equal to one. $\square$

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